

INDUSTRIAL PRODUCT DATASHEET

reComputer Rugged J4012 - Ruggedized IP66 Edge AI Computer with NVIDIA® Jetson Orin™ 16GB

The reComputer Rugged J4012 is a ruggedized IP66 edge AI computer with NVIDIA® Jetson Orin™ 16GB, M12 connectors, JetPack 5.1.3, and -20°C to 60°C operation for outdoor video analytics and autonomous logistics.

SKU

100046979

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PRICE

\$2299.00(Excl. VAT)



Overview



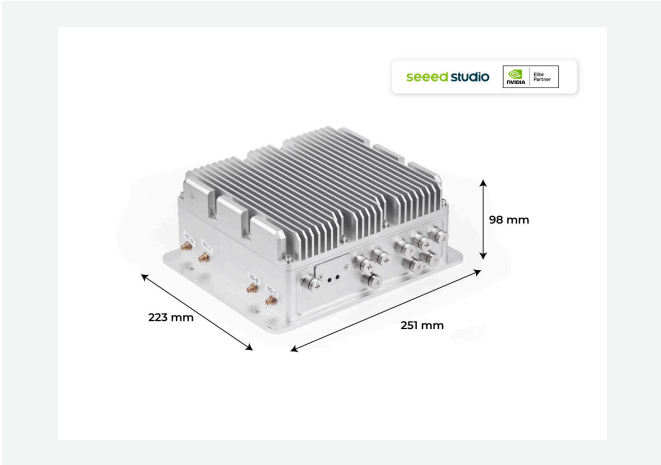
The reComputer Rugged J4012 is a ruggedized IP66 edge AI computer powered by NVIDIA® Jetson Orin™ 16GB, designed for harsh environments. It features lockable/waterproof M12 connectors for Ethernet (PSE), DI, DO, RS-232/422/485, and CAN, plus USB, comes pre-installed with JetPack 5.1.3 and WiFi6, and operates from -20°C to 60°C at 25W mode, supporting outdoor video analytics and autonomous logistics.

Key Features

- **Ruggedized Hardware Design:** Ruggedized edge AI computer with IP66 rating that can be deployed in harsh environment
- **Lockable and Waterproof connectors:** M12 for all interfaces (Ethernet (PSE), DI, DO, RS-232/422/485, and CAN), plus lockable/waterproof USB connectors.
- **Application and Benefit:** Ideal for rapid development of autonomous vehicles in harsh environments (AGV, AMR, trucks), accelerating time-to-market with ready-to-use interfaces and optimized AI frameworks.
- **Ready to Use:** Pre-installed with JetPack 5.1.3 and equipped with WiFi6 module.
- **Wide Operating Range:** Operates reliably across a temperature range of -20°C to 60°C at 25W mode

Product Gallery





Specification

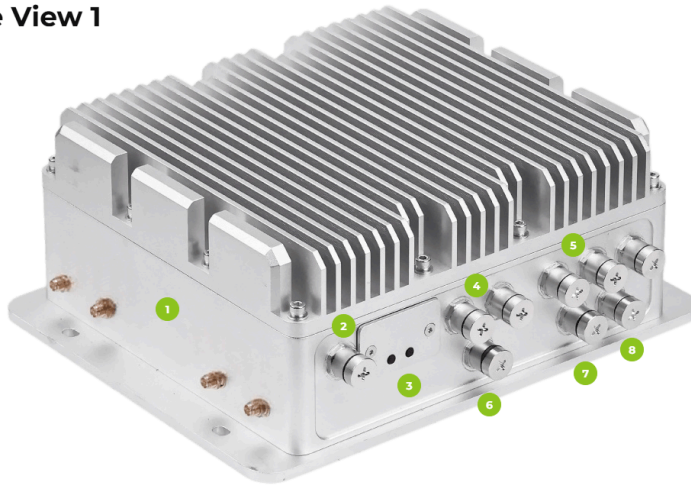
Module Specifications		
Product Name	reComputer Rugged J4012	
Module	NVIDIA Jetson Orin™ NX 16GB	
AI Performance	Orin NX 16GB – 100 TOPS	
GPU	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores	
GPU Max Frequency	918 MHz	
CPU	8-core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	
CPU Max Frequency	2.0 GHz	
DL Accelerator	2 x NVDLA v2.0	
DL Max Frequency	1.23 GHz	
Vision Accelerator	1 x PVA v2.0	
Memory	16GB 128-bit wide LPDDR5 DRAM 102.4GB/s	
Power	10W - 15W - 25W	
Video Encode	1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)	
Video Decode	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)	
Mechanical	69.6mm x 45mm 260-pin SO-DIMM connector	

Box Specifications		
Storage	1x M.2 KEY M PCIe (128G SSD included)	
Networking	M.2 KEY E	1x M.2 Key E for WiFi/Bluetooth module
	M.2 KEY B	1x M.2 Key B for 5G module
	SIM	1x Nano-SIM slot
	Ethernet	5x RJ-45 Gigabit Ethernet (4x PSE, all M12)

I/O	USB	4x USB 3.0 Type-A (5Gbps) 1x USB 3.0 Type-C (Recovery) 1x USB 2.0 Type-C (Debug)
	CAN	2x CAN (M12)
	Display	1x HDMI
	DI/DO	2x DI, 2x DO (M12)
	Serial	1x RS-232/422/485 (M12)
	RTC	1x RTC 2-pin 1x RTC Socket
	LED	2x LED (PWR, SSD)
	Button	1x Recovery 1x RESET
	Antenna Hole	4x Antenna Hole
Power	19-48V M12 A-code connector	
Jetpack Version	Jetpack 5.1.3	
Mechanical	Dimensions (W x D x H)	210mm*190mm*95mm (Bottom Panel is 250mm * 210mm)
	Weight	3.5kg
	Installation	Desk, Wall-mounting
Operating Temperature	-20°C~60°C (25W Mode)	
Warranty	2 Years	
Certification (Pending)	RoHS, CE, FCC	

Hardware Overview

Side View 1



- 1 Antenna x 4
- 2 DC 19~ 48 V
- 3 Reset and Recovery Button
USB Type-C x 2 (Debug+Recovery)
- 4 1 GbE x 1
- 5 1 GbE (PSE) x 4
- 6 CAN x 2
- 7 RS-232/422/485 x 1
- 8 DI x 2 DO x 2

Side View 2



- 1 Power LED
- 2 SSD LED
- 3 Vent Valve
- 4 USB 3.1 x 2
- 5 Display
- 6 USB 3.1 x 2

Applications

- **Outdoor video analytics:** For outdoor video analytics and monitoring, equipment must operate reliably outdoors despite constant exposure to UV radiation, moisture, temperature fluctuations, and mechanical vibration. With its IP66-rating enclosure, reComputer Rugged can be used in outdoor environments for such purposes. Paired with computing power of NVIDIA Jetson Orin modules, reComputer Rugged J40 is a powerful edge AI gateway for remote monitoring stations.
- **Autonomous Logistics:** Outdoor AMRs are increasingly deployed in urban and facility management environments to perform autonomous security patrols and last-mile delivery services. These multi-purpose robots are designed to navigate seamlessly through both indoor and outdoor environments. Therefore they need reComputer Rugged with NVIDIA® Jetson Orin™ as brain of AI-powered navigation systems that combine LiDAR, ultrasonic sensors, cameras, and GPS to map complex environments and avoid pedestrians, vehicles, and obstacles in real-time.

Part List

Jetson Orin™ NX 16GB module	x1
Seeed Carrier Board (reComputer Rugged J401)	x1
128GB NVMe SSD	x1
Aluminum Case and Heatsink	x1
Waterproof USB Cable; Type A to Type A	x1
M12 A-code Power Cable	x1
M12 A-code 8-pin Cable	x1
M12 X-code to RJ45 Cable	x1
Terminal Block female	x1
Terminal Block male	x1
DC Power cable	x1
USB Type-C to Type-A Cable	x1
User Manual	x1

Compliance & Logistics

HSCODE	8471419000
USHSCODE	8517180050
UPC	
EUHSCODE	8471707000
COO	CHINA

Resource Links

- <https://www.seeedstudio.com/reComputer-Rugged-J4012-p-6920.html>
- [User Manual & Datasheet](#)
- [Carrier Board Schematic](#)
- [PSE Board Schematic](#)
- [3D File](#)
- [Seeed NVIDIA Jetson Product Catalog](#)
- [Nvidia Jetson Comparison](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson One Pager](#)

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